Formal Methods in Software Engg. - Assignment 2 - Spin

1) a) Check if $G(Fp) \equiv F(Gp)$.

$G(Fp)$ means that p is true infinitely often. In other words, for any state, there exists a future state where p will be true. On the other hand, $F(Gp)$ means that at one point, p will be eternally true.

These two statements are not equivalent. The counter example for this is:

$$\varepsilon(p \sim p)^{\omega}$$

This implies an infinite sequence: $p, \sim p, p, \sim p, \ldots$

This sequence satisfies $G(Fp)$ but does not satisfy $F(Gp)$. Hence:

$$\boxed{G(Fp) \not\equiv F(Gp)}$$

b) Check if $p\,U\,q \equiv (p \vee q) \wedge G((p \wedge \sim q) \Rightarrow X(p \vee q))$

$p\,U\,q$ means that q should eventually happen, and p must hold until then. $(p \vee q) \wedge G((p \wedge \sim q) \Rightarrow X(p \vee q))$ means that we must start with p or q being true. Then, for all other subsequent states, whenever we have p but not q, the next state must have either p or q.

These two statements are not equivalent. The counter example for this is:

$$\varepsilon(p)^{\omega}$$

This implies an infinite sequence: $p, p, p \ldots$

This sequence does not satisfy $p\,U\,q$ but satisfies

$(p \vee q) \wedge G((p \wedge \sim q) \Rightarrow X(p \vee q))$. Hence:

$$\boxed{p \, U q \not\equiv (p \vee q) \wedge G((p \wedge \sim q) \Rightarrow X(p \vee q))}$$

2) The Büchi Automata is formally defined as:

$$A = (Q, S, \Delta, F),$$

where:

→ Q is a finite set of states
→ $S \subseteq Q$ is the set of initial states
→ $\Delta \subseteq Q \times A \times Q$ is the set of transitions
→ $F \subseteq Q$ is the set of final states

The language $L(A)$ of a Büchi automata consists of all infinite words that visit a final state infinitely often.

Hence, the complement, $L(\bar{A})$ must consist of words that visit F only finitely often. Therefore, to construct a Büchi automaton for $L(\bar{A})$, we must:

i) Visit the final states for a finite number of times.
ii) Transition to a state that will never visit the final state

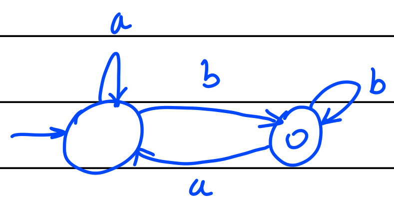
To do this, we construct a non-deterministic Büchi automaton B that simulates A. It then non-deterministically guesses a point after which no accepting state will be visited. After this, only non-accepting states can be seen.

We can do this using two modes: mode 0 implies that we have not made the guess yet. Mode 1 implies that the guess has been made, and that we must remain here forever.

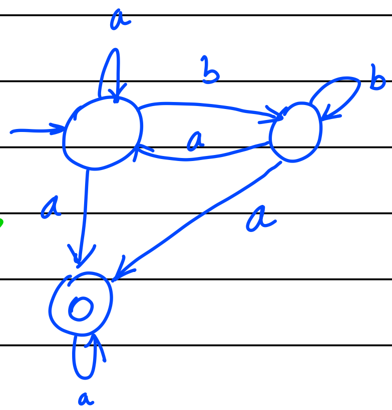
For each transition Δ of the form $q \xrightarrow{a} q'$, if we are in mode 0, we choose to stay in mode 0 or go to mode 1.

An example:

Büchi automaton for infinitely many b's (from slides)

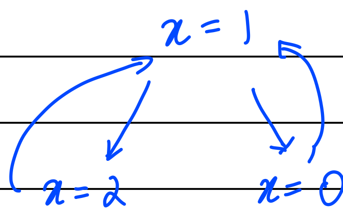


The complement automaton can be as follows:



This is where we make the guess

3) a) We can describe this system using a flowchart:

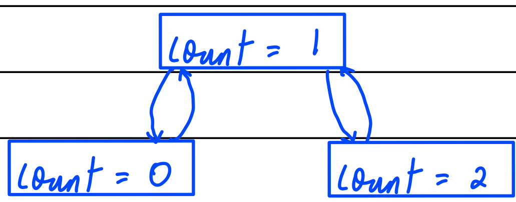


(initial state)

(both guards are true, any one will be chosen)

Hence, the system can be described by this sequence, 1, 0 or 2, 1, 0 or 2, ...

This can also be written as an automaton:



b) No. The model is satisfied by $(10)^\omega$, which represents the sequence count = 1, count = 0, count = 1, count = 0, ... count = 2 never occurs in this sequence. Hence, $x = 1 \not\Rightarrow F(x=2)$.

c) The formula $x = 1 \Rightarrow F(x=2)$ can be written (using p & q) as:

$$p \Rightarrow Fq$$

Let us obtain the negation of this formula:

$$\varphi = \sim(p \Rightarrow Fq) \equiv \sim(\sim p \vee Fq) \quad [a \rightarrow b \equiv \sim a \vee b]$$

$$\equiv \sim(\sim p) \wedge \sim(Fq) \quad [\sim(a \vee b) \equiv \sim a \wedge \sim b]$$

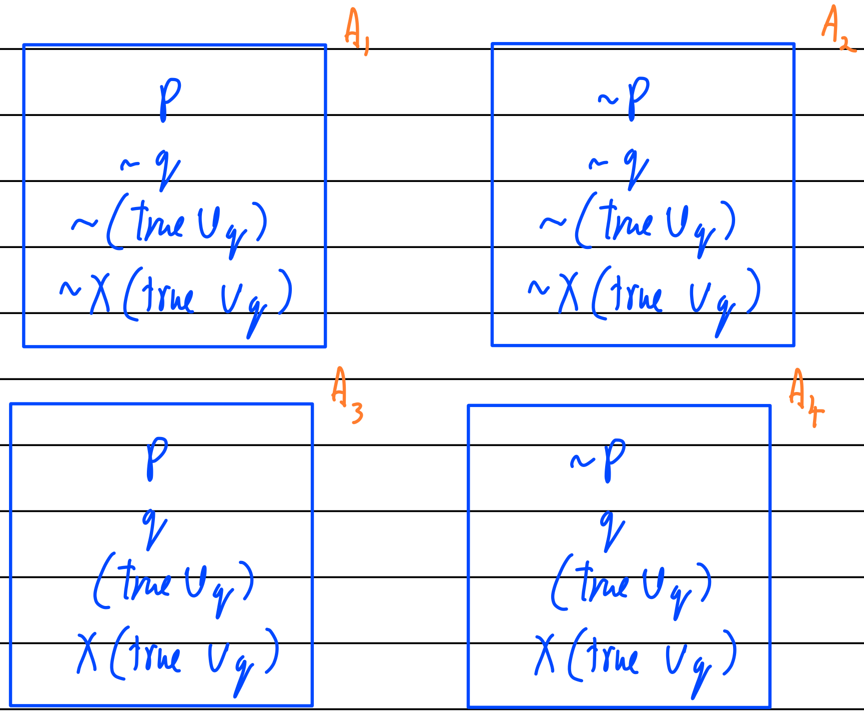
$$\equiv p \wedge \sim(\text{true}\ U\ q)$$

Let us now construct the Büchi automaton for $\varphi = p \wedge G(\sim q)$

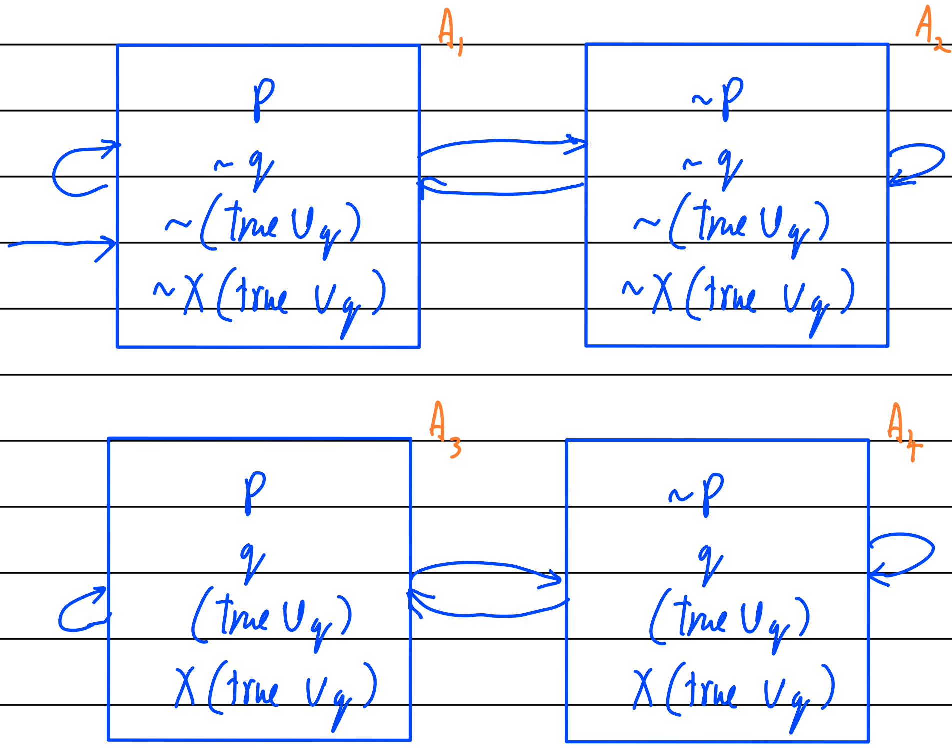
i) Obtain closure:

$$cl(\varphi) = \{p, \sim p, q, \sim q, \text{true}, \text{true}\ U\ q, \sim(\text{true}\ U\ q), X(\text{true}\ U\ q), \sim X(\text{true}\ U\ q), p \wedge \sim(\text{true}\ U\ q)\}$$

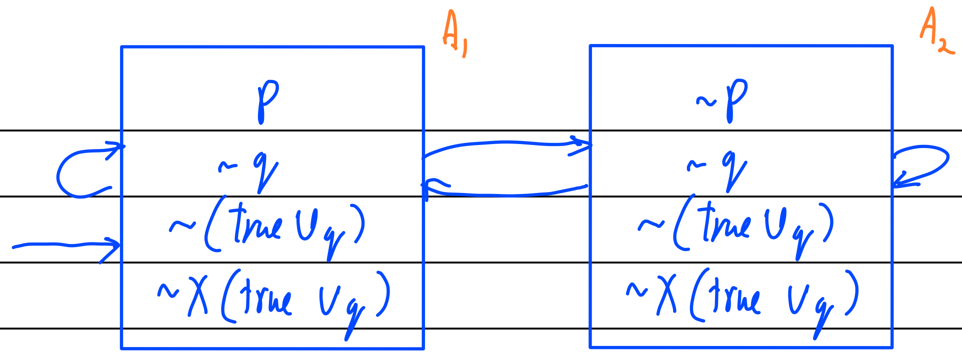
ii) Form atoms (states):



iii) Add transitions & initial state(s):

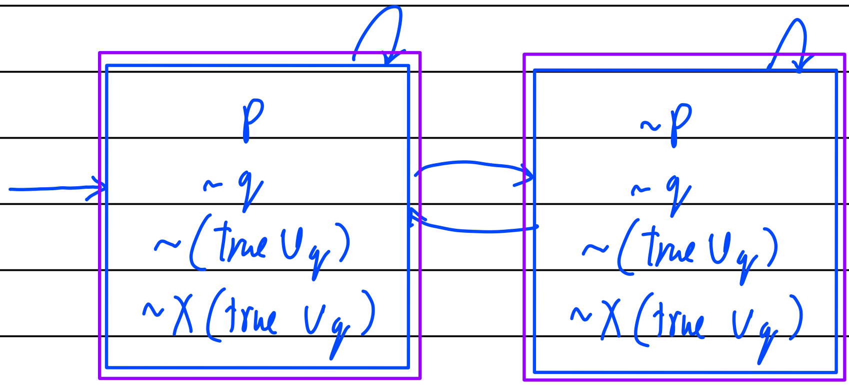


iv) The states A_3 & A_4 are unreachable. Hence, we can 'prune' them:

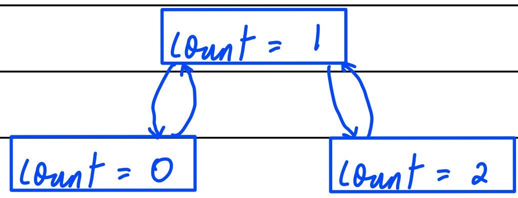


v) Add accepting states (both states are accepting)

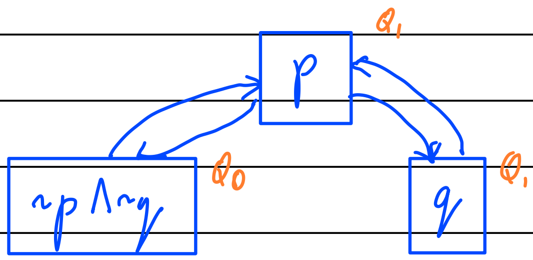
$$F = \{A \mid (\text{true}\ U\ q) \notin A \ \vee\ q \in A\}$$



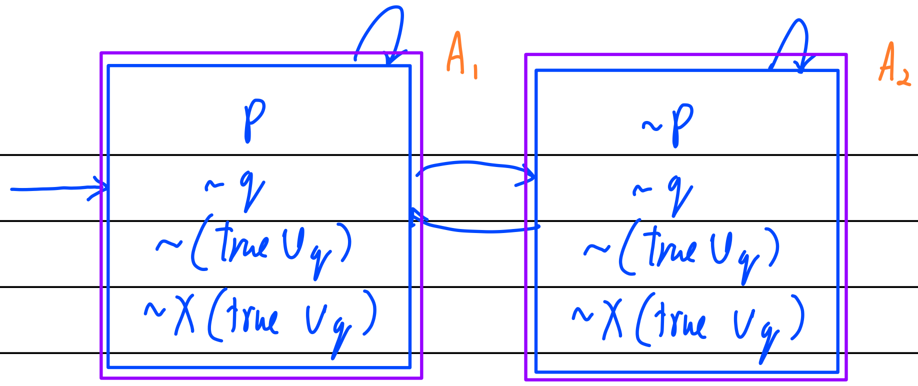
d) The transition system is as follows:



We can re-write this in terms of p & q (for compatibility with the Büchi automata:



The Büchi automaton is as follows:



Let us identify the products (so that we can identify which cells are possible):

$Q_0 \times A_1$: ✗

$\sim p$
$\sim q$
p
$\sim q$
$\sim(\text{true}\ U\ q)$
$\sim X(\text{true}\ U\ q)$

$Q_0 \times A_2$: ✓

$\sim p$
$\sim q$
$\sim p$
$\sim q$
$\sim(\text{true}\ U\ q)$
$\sim X(\text{true}\ U\ q)$

$Q_1 \times A_1$: ✓

p
p
$\sim q$
$\sim(\text{true}\ U\ q)$
$\sim X(\text{true}\ U\ q)$

$Q_1 \times A_2$: ✗

p
$\sim p$
$\sim q$
$\sim(\text{true}\ U\ q)$
$\sim X(\text{true}\ U\ q)$

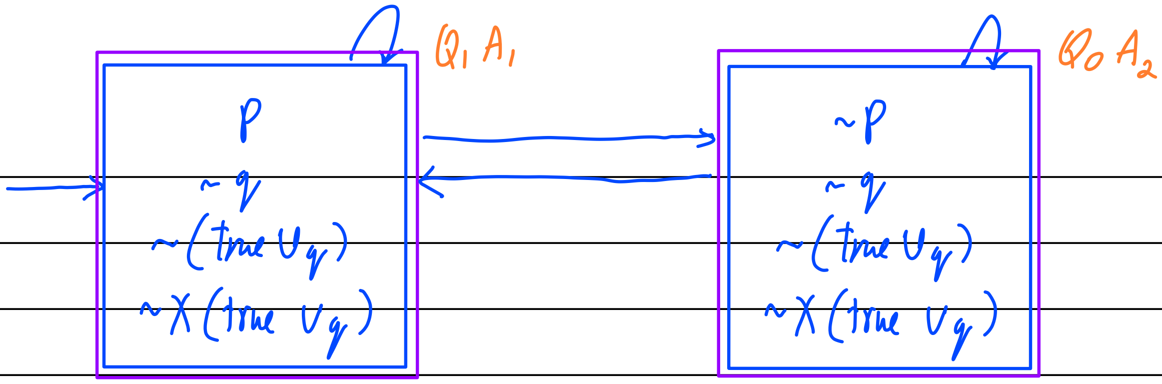
$Q_2 \times A_1$: ✗

q
p
$\sim q$
$\sim(\text{true}\ U\ q)$
$\sim X(\text{true}\ U\ q)$

$Q_2 \times A_2$: ✗

q
$\sim p$
$\sim q$
$\sim(\text{true}\ U\ q)$
$\sim X(\text{true}\ U\ q)$

Hence, the only possible states are $Q_0 \times A_2$ and $Q_1 \times A_1$. Let us construct the automaton:



4) Let $\varphi = \text{true}\ U\ p$

i) Obtaining the closure:

$$cl(\varphi) = \{p, \sim p, \text{true}, \text{true}\ U\ p, \sim(\text{true}\ U\ p), X(\text{true}\ U\ p), \sim X(\text{true}\ U\ p)\}$$

ii) Form Atoms:

A_1: p, $\text{true}\ U\ p$, $X(\text{true}\ U\ p)$

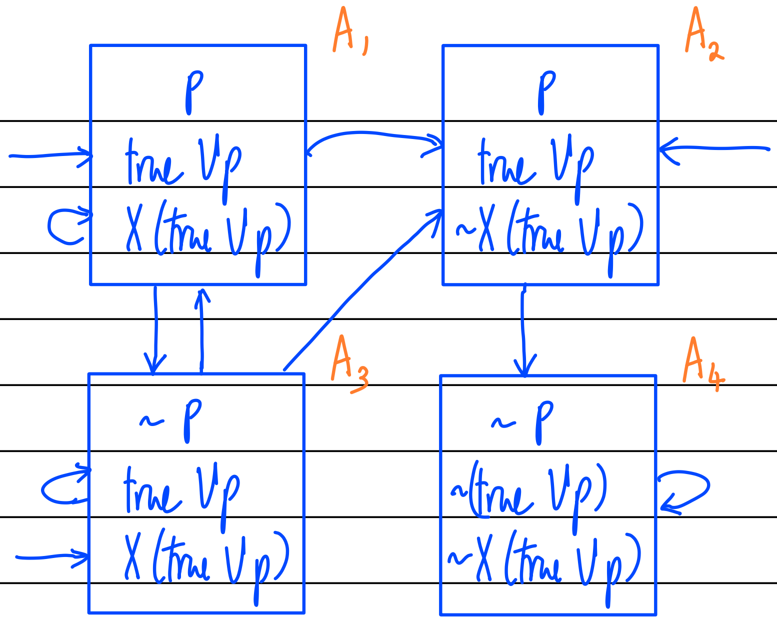
A_2: p, $\text{true}\ U\ p$, $\sim X(\text{true}\ U\ p)$

A_3: $\sim p$, $\text{true}\ U\ p$, $X(\text{true}\ U\ p)$

A_4: $\sim p$, $\sim(\text{true}\ U\ p)$, $\sim X(\text{true}\ U\ p)$

iii) Add transitions & initial states:

- A_1, A_2 & A_3 are all initial because they contain φ.



iv) Add final states : $F_{ov2} = \{ A \mid true \ U \ p \notin A \ \vee \ p \in A \}$

Hence, A_1, A_2 & A_4 are final states: